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Evaluating Sakana's AI Scientist for Autonomous Research: Wishful Thinking or an Emerging Reality Towards 'Artificial Research Intelligence' (ARI)?

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Abstract. Recently, Sakana.ai introduced the AI Scientist, a system claiming to automate the entire research lifecycle and conduct research autonomously, a concept we term *Artificial Research Intelligence* (ARI). Achieving ARI would be a major milestone toward Artificial General Intelligence (AGI) and a prerequisite to achieving Super Intelligence. The AI Scientist received much attention in the academic and broader AI community. A thorough evaluation of the AI Scientist, however, had not yet been conducted.

We evaluated the AI Scientist and found several critical shortcomings. The system's literature review process is inadequate, relying on simplistic keyword searches rather than profound synthesis, which leads to poor novelty assessments. In our experiments, several generated research ideas were incorrectly classified as novel, including well-established concepts such as micro-batching for stochastic gradient descent (SGD). The AI Scientist also lacks robustness in experiment execution—five out of twelve proposed experiments (42%) failed due to coding errors, and those that did run often produced logically flawed or misleading results. In one case, an experiment designed to optimize energy efficiency reported improvements in accuracy while consuming more computational resources, contradicting its stated goal. Furthermore, the system modifies experimental code minimally, with each iteration adding only 8% more characters on average, suggesting limited adaptability. The generated manuscripts were poorly substantiated, with a median of just five citations per paper—most of which were outdated (only five out of 34 citations were from 2020 or later). Structural errors were frequent, including missing figures, repeated sections, and placeholder text such as "Conclusions Here". Hallucinated numerical results were contained in several manuscripts, undermining the reliability of its outputs.

Despite its limitations, the AI Scientist represents a significant leap forward in research automation. It produces complete research manuscripts with minimal human intervention, challenging conventional expectations of AI-generated scientific work. Many reviewers or university instructors conducting only a superficial assessment may struggle to distinguish its output from that of human researchers, demonstrating how far AI has progressed in mimicking academic writing and structuring scientific arguments. While the quality of its manuscripts currently aligns with that of an unmotivated undergraduate student rushing to meet a deadline, this level of autonomy in research generation is remarkable. More strikingly, it achieves this at an unprecedented speed and cost efficiency—our analysis indicates that generating a full research paper costs only $6–$15, with just 3.5 hours of human involvement. This is significantly faster than traditional human researchers. Given that AI research automation was nearly nonexistent just a few years ago, the AI Scientist marks a substantial milestone toward Artificial Research Intelligence (ARI), signalling the acceleration of AI-driven scientific discovery.

The AI Scientist also illustrates the urgent need for a discussion within the Information Retrieval (IR) and broader scientific communities. Whether and when ARI becomes a reality depends on how the academic and AI communities shape its development and governance. We propose concrete steps, including pilot projects and competitions, and standardized attribution frameworks such as research logs and markup languages.

CCS Concepts: • **Computing methodologies** → **Machine learning**; **Artificial intelligence**; • **Information systems** → **Language models**.

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1 Introduction

In autumn 2024, Sakana.ai, a Tokyo-based start-up that has raised $200 million in funding[1], announced the "AI Scientist". With the AI Scientist, Sakana boldly promised the "beginning of a new era in scientific discovery" [12]. The open-source[2] AI Scientist is supposed to "automate the entire research lifecycle"; i.e. it generates research ideas, designs and conducts experiments, analyzes results, writes research papers, and finally even reviews them — essentially automating the daily work of millions of researchers worldwide. According to Sakana, the AI Scientist produces research papers for approximately $15 each. Sakana acknowledges "occasional flaws" and explains further limitations in a pre-print manuscript [8]. Yet, based on Sakana's released information, most readers will understand that the AI Scientist appears to be a fully-functional system. Especially considering Sakana's claims that "It is worth noting that the [AI Scientist] can autonomously run the entire life cycle of machine learning research without any human intervention except for initial preparation" [13] and that the peer review system works "with near-human accuracy" [12].

Sakana's AI Scientist is not the only AI tool that aims to support or even replace significant aspects of scientific work. A rapidly expanding body of research explores AI's role in scientific discovery, with recent preprints examining its impact on literature retrieval, idea generation, and automated experimental design. Several studies indicate that large language models (LLMs) can already generate research ideas comparable to those of human scientists [15, 18, 19], with some findings suggesting AI may even surpass human creativity [11, 14]. Google has framed AI as ushering in a "new era of scientific discovery" [10], while specialized workshops such as "Towards Agentic AI for Science" at ICLR 2025 highlight the field's growing momentum[3].

Notably, just one day before the submission of our current manuscript for review to a major IR conference, and hours before uploading it to arXiv, Google announced the *AI Co-Scientist*[4], indicating the emerging interest in autonomous research agents, or, as we term it 'Artificial Research Intelligence' (ARI), i.e. artificial intelligence that has not yet reached the level of general intelligence, but intelligence that is capable of conducting research that is indistinguishable from human research. In this context it is also interesting to note that [7] submitted a paper on AI-driven idea generation to ICLR 2025, which was rejected after an intense review process involving 45 discussion posts — described by the conference chair as "filled with drama"[5]. This case exemplifies the unease among researchers and reviewers who face the prospect of AI encroaching on their roles, raising questions about the future of human-led scientific inquiry.

Despite the large body of work, the AI Scientist is probably the tool that has made the boldest promises, yet. And, possibly because of this, it has gained the most attention, both within the academic community and among the broader AI enthusiast audience. As of early February 2025, Sakana's pre-print manuscript gained more than 100 citations[6], their

[1] https://sakana.ai/series-a/
[2] https://github.com/SakanaAI/AI-Scientist
[3] https://blog.iclr.cc/2025/02/03/workshops-at-iclr-2025
[4] https://blog.google/feed/google-research-ai-co-scientist/
[5] https://openreview.net/forum?id=GHJzxPgFa6&noteId=tN3UypVtcw
[6] https://scholar.google.com/scholar?cites=3682611450429091902

GitHub repository received 8.9k stars and it was forked over 1.3k times. Prominent AI content creators on YouTube and X, each with hundreds of thousands of followers [5, 16], enthusiastically discuss the AI Scientist.

Notably, the most positive reviewers of the AI Scientist, e.g., [5], have not tested the system themselves but rely solely on Sakana's information. Meanwhile, users in online forums and the GitHub issue trackers report difficulties in setting up the AI Scientist[7] or encountering technical issues[8]. Other discussions[9] use the AI Scientist's release as a broader reflection on AI's role in scientific discovery. Some reviews provide in-depth analyses[10], examining its source code and generated papers. However, to the best of our knowledge, a comprehensive, independent assessment based on direct experimentation has yet to be conducted.

The AI Scientist is a machine learning driven system, making it highly relevant to the machine learning research community. However, its impact extends beyond machine learning, with profound implications for the Information Retrieval (IR) community, which motivates us to write this paper. While a recent SIGIR perspective paper [20] provided valuable insights into how LLMs are transforming traditional IR tasks, we must also consider a broader question: What happens when LLMs move beyond assisting IR research and begin to conduct it autonomously? This possibility is no longer theoretical — AI systems are increasingly capable of generating research ideas, designing experiments, and even evaluating scientific work.

In this paper, we take the next step in this discussion by providing an independent, systematic assessment of the AI Scientist's capabilities, limitations, and trajectory. Our goal is not only to evaluate its functionality but to highlight its potential role in reshaping scientific inquiry, particularly within information retrieval. By analyzing its performance across idea generation, experimentation, and manuscript production and review, we offer a comprehensive perspective on what AI-driven research tools can — and cannot — achieve today and may achieve in the future.

Our study focuses exclusively on Sakana's AI Scientist rather than other comparable tools due to its particularly bold claims and the significant attention it has garnered within the research community. While numerous AI-driven tools support various aspects of scientific research—including AI-assisted literature retrieval and summarization (Semantic Scholar, Elicit, Scite.ai), automated experiment design (IBM's Watson Discovery), and research paper generation — the AI Scientist stands out for its promise to fully automate the entire research lifecycle, from idea generation to peer review. This ambitious goal, combined with its open-source nature and extensive media coverage, makes it a uniquely compelling subject for independent evaluation. Other tools, while valuable in their respective domains, typically focus on specific tasks rather than claiming end-to-end automation of scientific inquiry. By concentrating on the AI Scientist, we aim to provide a detailed and critical analysis of its capabilities and limitations without diluting our focus across multiple tools with varying objectives and architectures.

Our findings reveal a system that, while far from replacing human researchers, demonstrates the potential to automate significant parts of the research process. The AI Scientist does not yet fulfill its promises, struggling with methodological soundness, experimental execution, and literature retrieval. However, these limitations are technical hurdles rather than fundamental barriers — challenges that will likely be addressed as AI-driven research systems evolve.

For the IR community, the emergence of such tools presents both an opportunity and a challenge. On one hand, they offer new ways to automate literature retrieval, citation analysis, and experimental design, potentially advancing the field. On the other, they raise critical questions about the role of human researchers, the nature of scientific discovery, and the reliability and generalizability of AI-generated knowledge.

[7] https://www.reddit.com/r/learnmachinelearning/comments/1fuw2yb/has_anybody_gotten_sakana_ai_scientist_to_work/
[8] https://github.com/SakanaAI/AI-Scientist/issues
[9] https://spectrum.ieee.org/ai-for-science-2
[10] https://medium.com/@nimritakoul01/evaluating-the-ai-scientist-63e419e575b8